PROJECT REPORT
GitHub Profile Portfolio

A curated GitHub profile repository that presents the developer identity, technical skills, and featured projects of Tejas Gowda M (tejas448866).

Project Title	GitHub Profile Portfolio — tejas448866
Student / Owner	Tejas Gowda M
Repository	github.com/tejas448866/tejas448866
Type	Special GitHub profile README repository

1. Project Purpose

The repository **tejas448866/tejas448866** is a special GitHub profile repository. When a user creates a repository with the same name as their username, GitHub automatically renders its README.md on the public profile page. This project uses that mechanism to present a professional developer identity, technical skill set, and a curated showcase of featured work.

The profile positions the owner as an **AI Systems Engineer & Full-Stack Developer** with focus areas in Computer Vision and 3D Interactive Systems. It serves as a living portfolio that visitors, recruiters, and collaborators can use to understand the developer's interests, stack, and major projects at a glance.

Target audience includes fellow developers, academic evaluators, potential employers, and anyone exploring open-source work in real-time computer vision, autonomous agents, and interactive systems.

2. Technologies Used

The profile repository itself is primarily Markdown and SVG. The skills and stacks it advertises (and that underpin the featured projects) are:

Languages

- Python — primary language for AI, vision, and data projects
- TypeScript / JavaScript — front-end, 3D, and optimization tools
- HTML5 / CSS3 — web presentation
- Jupyter — exploratory data analysis and notebooks

AI & Vision

- OpenCV — image and video processing
- MediaPipe — real-time hand and pose landmark detection
- Computer Vision pipelines and autonomous agent patterns

Development & Tools

- Node.js — server-side JavaScript runtime
- Three.js / 3D engines — interactive 3D experiences
- Git & GitHub — version control and profile hosting
- SVG assets — light/dark mode profile header graphics

3. Working and Structure

Repository Contents

The repository is intentionally minimal. It contains:

- **README.md** — the full profile document rendered on the GitHub profile page
- **dark_mode.svg** / **light_mode.svg** — responsive header graphics that adapt to the viewer's colour scheme

Profile Sections

1. **Header & Identity** — name, role title, and follower/repo badges.
2. **About Me** — short professional summary and current focus areas.
3. **Tech Stack & Skills** — categorized badges for Languages, AI & Vision, and Development.
4. **Featured Projects** — six highlighted repositories with short descriptions and stack badges.
5. **GitHub Activity & Statistics** — dynamic stats, top languages, and streak widgets.
6. **Connect** — links back to the profile and repositories.

How It Works

GitHub detects a repository whose name matches the account username and treats its README as the profile landing page. The Markdown uses HTML `<picture>` elements so the SVG header switches between dark and light mode automatically. External services (github-readme-stats, streak stats) inject live activity graphics. All content is static Markdown; no build step or backend is required for the profile itself.

4. Featured Projects

The profile showcases six projects that represent the owner's main technical themes:

■ **PalmPilot-AI — https://github.com/tejas448866/PalmPilot-AI**
Real-time hand-gesture gaming controller. A webcam + MediaPipe pipeline converts hand movements into keyboard actions for browser racing games, without a physical controller.

■ **AI-VIRTUAL-ASSISTANT — https://github.com/tejas448866/AI-VIRTUAL-ASSISTANT**
Autonomous desktop assistant with system monitoring, vision-based screen inspection, automated browsing, code debugging, and task automation.

■■ **3D Stealth Horror Game — https://github.com/tejas448866/3d-stealth-horror-game-1-**
Immersive 3D survival experience: navigate abandoned environments, solve environmental puzzles, avoid patrol enemies, and perform stealth escapes.

■ **Smart Timetable Optimization — https://github.com/tejas448866/Smart-Timetable-Optimization**
Constraint-optimisation engine that produces conflict-free schedules across instructors, classrooms, student batches, and curriculum rules.

■ **Financial Decision Engine — https://github.com/tejas448866/artifacts**
AI-based business and financial decision engine offering structured constraint analysis and predictive optimisation models.

■ **VoltRelay Data Analysis — https://github.com/tejas448866/VoltRelay-Data-Analysis**
Exploratory data analysis, statistical modelling, and telemetry trend visualisation using Jupyter notebooks and data-science libraries.

5. Testing, Limitations and Maintenance

As a static Markdown + SVG profile, traditional unit testing does not apply. Quality is maintained by visual review across light/dark themes and by keeping project links and descriptions accurate.

Limitations

- The repository does not contain application source code; it only links to other projects.
- Dynamic stats widgets depend on third-party services remaining available.
- Profile content must be updated manually when new projects are added or descriptions change.
- SVG header assets must stay in sync with any branding changes.

Possible Future Improvements

- Expand the featured-project set with newer work and clearer impact metrics.
- Add a short "Selected contributions" or publications section if applicable.
- Provide a lightweight personal site or portfolio page that mirrors the same content.

- Keep README structure consistent so visitors can scan projects quickly.

6. GitHub and Originality

The repository is the canonical GitHub profile page for the account **tejas448866**. All project descriptions and links point to the owner's public repositories. The profile text, structure, and SVG assets are original content created for this account.

Execution / Viewing Steps

1. Open https://github.com/tejas448866 in any browser.
2. The profile README renders automatically from the tejas448866/tejas448866 repository.
3. Click any featured-project link to visit the corresponding source repository.
4. To edit: clone the repo, update README.md or the SVG files, and push to the main branch.

Individual Contributions

I, Tejas Gowda M, created and maintain this profile repository, including the About section, skill presentation, featured-project selection and descriptions, and the light/dark mode header assets.

Learning Outcomes

- Using GitHub's special profile-repository convention to build a professional online presence.
- Structuring Markdown and HTML for readable, scannable technical portfolios.
- Presenting computer-vision, agent, 3D, and optimisation work in a consistent format.
- Maintaining a single source of truth that links to multiple technical projects.

Conclusion

The tejas448866/tejas448866 repository successfully serves as a focused developer portfolio on GitHub. By combining a clear professional identity, a well-organised tech-stack display, and six featured projects spanning real-time computer vision (PalmPilot-AI), autonomous assistance, 3D interactive systems, constraint optimisation, financial decision modelling, and data analysis, the profile gives visitors an immediate and accurate picture of the owner's technical interests and capabilities. The minimal repository structure keeps maintenance simple while remaining fully functional as a living portfolio.

— End of Report —